

AI Task – 1

1. Simple Reflex Agent Vacuum Cleaner

Code:

```
rooms = {  
    "A": "Dirty",  
    "B": "Clean"  
}  
  
location = 'A'  
  
if rooms[location] == "Dirty":  
    rooms[location] = "Clean"  
    print("Cleaning room")  
else:  
    location = "B"  
    print("Agent moves to another room")
```

Output(location = 'A'):

Cleaning room

Output(location = 'B'):

Agent moves to another room

2. Model-Based Reflex Agent Vacuum Cleaner

Code:

```
rooms = {  
    "A": "Dirty",  
    "B": "Dirty",  
    "C": "Clean"  
}
```

```

for room in rooms:
    print("Current room:", room)
    if rooms[room] == "Dirty":
        print("Cleaning Room:", room)
        rooms[room] = "Clean"
print(rooms)

```

Output:

```

Current room: A
Cleaning Room: A
Current room: B
Cleaning Room: B
Current room: C
{'A': 'Clean', 'B': 'Clean', 'C': 'Clean'}

```

3. Goal-Based Agent Vacuum Cleaner

Code:

```

def vacuum_agent(room_A, room_B, position):

    # Goal: Both the rooms should be clean
    if room_A == 0 and room_B == 0:
        return "Goal Achived - Both rooms are clean"

    # If current room is dirty, clean it
    if position == "A" and room_A == 1:
        return "Suck"

    if position == "B" and room_B == 1:
        return "Suck"

    # Move to the other room
    if position == "A":
        return "Right"
    if position == "B":
        return "Left"

```

```
# Initial state
room_A = 1
room_B = 1
position = "A"

print("Initial state")
print("Room A:", room_A)
print("Room B:", room_B)
print("Position:", position)

action = vacuum_agent(room_A, room_B, position)
print("Action:", action)

if action == "Suck":
    if position == "A":
        room_A = 0
    else:
        room_B = 0

elif action == "Right":
    position = "B"
elif action == "Left":
    position = "A"

print("\nFinal State:")
print("Room A:", room_A)
print("Room B:", room_B)
print("Goal Achieved!")
```

Output 1:

```
Initial state
Room A: 1
Room B: 1
Position: A
Action: Suck
```

Final State:

Room A: 0
Room B: 1
Goal Achieved!

Output 2:

Initial state
Room A: 1
Room B: 0
Position: B
Action: Left

Final State:
Room A: 1
Room B: 0
Goal Achieved!

4. Utility-Based Agent Vacuum Cleaner

Code:

```
def vacuum_agent(room_A, room_B, position, actions):  
    if position == "A":  
        if room_A == 1:  
            action = "Suck"  
            actions["Suck"] = 10  
        else:  
            action = "Right"  
            actions["Right"] = -1  
    else:  
        if room_B == 1:  
            action = "Suck"  
            actions["Suck"] = 10  
        else:  
            action = "Left"  
            actions["Left"] = -1  
    return action
```

```

actions = {"Suck": 0, "Right": 0, "Left": 0}

# Initial state
room_A = 1
room_B = 1
position = "A"

print("Initial state")
print("Room A:", room_A)
print("Room B:", room_B)
print("Position:", position)

# First action
action = vacuum_agent(room_A, room_B, position, actions)
print("Action:", action)

# Perform action
if action == "Suck":
    if position == "A":
        room_A = 0
    else:
        room_B = 0

elif action == "Right":
    position = "B"

elif action == "Left":
    position = "A"

print("\nFinal State:")
print("Room A:", room_A)
print("Room B:", room_B)
print("Position:", position)
print("Actions taken:", actions)

```

Output 1:

```

Initial state
Room A: 1

```

Room B: 1
Position: A
Action: Suck

Final State:
Room A: 0
Room B: 1
Position: A
Actions taken: {'Suck': 10, 'Right': 0, 'Left': 0}

Output 2:

Initial state
Room A: 1
Room B: 0
Position: B
Action: Left

Final State:
Room A: 1
Room B: 0
Position: A
Actions taken: {'Suck': 0, 'Right': 0, 'Left': -1}